Activity using SQLite Database

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COURSE CODE:DSA0503
COURSE NAME:QUERY PROCESSING

Question 1: Create the Department Table (Parent Table):

Algorithm:

1. Import the required libraries (sqlite3, pandas, and tabulate).
2. Create or connect to the SQLite database (college.db).
3. Create a cursor object.
4. Create the Department table with Department_ID as the Primary Key and Department_Name as the Text field.
5. Delete existing records (if any) from the Department table.
6. Insert the department records (CSE, ECE, EEE, MECH).
7. Save the changes using commit().
8. Retrieve all records from the Department table.
9. Display the table in row and column format with borders.
10. Close the database connection.

Code:

```
!pip install tabulate

import sqlite3
import pandas as pd
from tabulate import tabulate

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS Department(
    Department_ID INTEGER PRIMARY KEY,
    Department_Name TEXT
)
""")

cursor.execute("DELETE FROM Department")

department_data = [
    (1, "CSE"),
```

```
    (2, "ECE"),
    (3, "EEE"),
    (4, "MECH")
]

cursor.executemany(
    "INSERT INTO Department (Department_ID, Department_Name) VALUES (?, ?)",
    department_data
)

conn.commit()

df = pd.read_sql_query("SELECT * FROM Department", conn)

print("Department Table")
print(tabulate(df, headers="keys", tablefmt="grid", showindex=False))

conn.close()
```

Output:

```
Requirement already satisfied: tabulate in /usr/local/lib/python3.12/d
Department Table
+-----------------+-------------------+
|   Department_ID | Department_Name   |
+=================+===================+
|               1 | CSE               |
+-----------------+-------------------+
|               2 | ECE               |
+-----------------+-------------------+
|               3 | EEE               |
+-----------------+-------------------+
|               4 | MECH              |
+-----------------+-------------------+
```

Question 2: Create the Student Table (Child Table):

Algorithm:

1. Import the required libraries (sqlite3, pandas, and tabulate).
2. Connect to the SQLite database (college.db).
3. Enable foreign key constraints.
4. Create the Student table with the required attributes.
5. Delete existing records (if any) from the Student table.
6. Insert sample student records.
7. Save the changes using commit().
8. Retrieve all records from the Student table.

9. Display the records in a table with rows and columns.
10. Close the database connection.

Code:

```
!pip install tabulate

import sqlite3
import pandas as pd
from tabulate import tabulate

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("PRAGMA foreign_keys = ON")

cursor.execute("""
CREATE TABLE IF NOT EXISTS Student(
   Student_ID INTEGER PRIMARY KEY,
   Name TEXT,
   Age INTEGER,
   Department_ID INTEGER,
   CGPA REAL,
   FOREIGN KEY (Department_ID)
   REFERENCES Department(Department_ID)
)
""")

cursor.execute("DELETE FROM Student")

student_data = [
   (101, "Arun", 20, 1, 8.9),
   (102, "Priya", 21, 2, 7.8),
   (103, "Rahul", 20, 1, 9.2),
   (104, "Sneha", 22, 3, 8.4),
   (105, "Kiran", 21, 4, 7.5)
]

cursor.executemany(
   "INSERT INTO Student VALUES (?, ?, ?, ?, ?)",
   student_data
)

conn.commit()

df = pd.read_sql_query("SELECT * FROM Student", conn)

print("Student Table")
print(tabulate(df, headers="keys", tablefmt="grid", showindex=False))
```

conn.close()

Output:

```
Requirement already satisfied: tabulate in /usr/local/lib/python3
Student Table
+--------------+--------+-------+-----------------+--------+
|   Student_ID | Name   |   Age |   Department_ID |   CGPA |
+==============+========+=======+=================+========+
|          101 | Arun   |    20 |               1 |    8.9 |
+--------------+--------+-------+-----------------+--------+
|          102 | Priya  |    21 |               2 |    7.8 |
+--------------+--------+-------+-----------------+--------+
|          103 | Rahul  |    20 |               1 |    9.2 |
+--------------+--------+-------+-----------------+--------+
|          104 | Sneha  |    22 |               3 |    8.4 |
+--------------+--------+-------+-----------------+--------+
|          105 | Kiran  |    21 |               4 |    7.5 |
+--------------+--------+-------+-----------------+--------+
```

Question 3: Display Students from CSE Department:

Algorithm:

1. Import the required libraries (sqlite3, pandas, and tabulate).
2. Connect to the SQLite database (college.db).
3. Write an SQL query to retrieve students whose department is **CSE**.
4. Execute the query.
5. Store the result in a Pandas DataFrame.
6. Display the records in a table with rows and columns.
7. Close the database connection.

Code:

```
!pip install tabulate

import sqlite3
import pandas as pd
from tabulate import tabulate

conn = sqlite3.connect("college.db")

query = """
SELECT Student_ID, Name, Age, CGPA
FROM Student
WHERE Department_ID = (
    SELECT Department_ID
    FROM Department
```

```
    WHERE Department_Name = 'CSE'
)
"""

df = pd.read_sql_query(query, conn)

print("Students from CSE Department")
print(tabulate(df, headers="keys", tablefmt="grid", showindex=False))

conn.close()
```

Output:

```
Students from CSE Department
+--------------+--------+-------+--------+
|   Student_ID | Name   |   Age |   CGPA |
+==============+========+=======+========+
|          101 | Arun   |    20 |    8.9 |
+--------------+--------+-------+--------+
|          103 | Rahul  |    20 |    9.2 |
+--------------+--------+-------+--------+
```

Question 4: Count Students in Each Department:

Algorithm:

1. Import the required libraries (sqlite3, pandas, and tabulate).
2. Connect to the SQLite database (college.db).
3. Write an SQL query using **LEFT JOIN** to combine the Department and Student tables.
4. Use the COUNT() function to count the number of students in each department.
5. Group the results by department name using GROUP BY.
6. Display the result in a table with rows and columns.
7. Close the database connection.

Code:

```
!pip install tabulate

import sqlite3
import pandas as pd
from tabulate import tabulate

conn = sqlite3.connect("college.db")

query = """
SELECT Department.Department_Name,
COUNT(Student.Student_ID) AS Total_Students
```

```
FROM Department
LEFT JOIN Student
ON Department.Department_ID = Student.Department_ID
GROUP BY Department.Department_Name
"""

df = pd.read_sql_query(query, conn)

print("Count of Students in Each Department")
print(tabulate(df, headers="keys", tablefmt="grid", showindex=False))

conn.close()
```

Output:

```
Count of Students in Each Department
+-------------------+------------------+
| Department_Name   |   Total_Students |
+===================+==================+
| CSE               |                2 |
+-------------------+------------------+
| ECE               |                1 |
+-------------------+------------------+
| EEE               |                1 |
+-------------------+------------------+
| MECH              |                1 |
+-------------------+------------------+
```

Question 5: Update the Foreign Key (Change the Department of a Student):

Algorithm:

1. Import the required libraries (sqlite3, pandas, and tabulate).
2. Connect to the SQLite database (college.db).
3. Create a cursor object.
4. Update the Department_ID of the selected student using the UPDATE statement.
5. Save the changes using commit().
6. Retrieve the updated student records.
7. Display the records in a table with rows and columns.
8. Close the database connection.

Code:

```
!pip install tabulate
```

```
import sqlite3
import pandas as pd
from tabulate import tabulate

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
UPDATE Student
SET Department_ID = 3
WHERE Student_ID = 102
""")

conn.commit()

query = """
SELECT Student.Student_ID,
Student.Name,
Department.Department_Name,
Student.CGPA
FROM Student
JOIN Department
ON Student.Department_ID = Department.Department_ID
"""

df = pd.read_sql_query(query, conn)

print("Student Table After Updating Department")
print(tabulate(df, headers="keys", tablefmt="grid", showindex=False))

conn.close()
```

Output:

```
Student Table After Updating Department
+--------------+--------+-------------------+--------+
|   Student_ID | Name   | Department_Name   |   CGPA |
+==============+========+===================+========+
|          101 | Arun   | CSE               |    8.9 |
+--------------+--------+-------------------+--------+
|          102 | Priya  | EEE               |    7.8 |
+--------------+--------+-------------------+--------+
|          103 | Rahul  | CSE               |    9.2 |
+--------------+--------+-------------------+--------+
|          104 | Sneha  | EEE               |    8.4 |
+--------------+--------+-------------------+--------+
|          105 | Kiran  | MECH              |    7.5 |
+--------------+--------+-------------------+--------+
```

Question 6: Delete a Department:

Algorithm:

1. Import the required libraries (sqlite3, pandas, and tabulate).
2. Connect to the SQLite database (college.db).
3. Create a cursor object.
4. Delete the student records belonging to the selected department.
5. Delete the department from the Department table.
6. Save the changes using commit().
7. Retrieve the updated Department table.
8. Display the table in rows and columns.
9. Close the database connection.

Code:

```
!pip install tabulate

import sqlite3
import pandas as pd
from tabulate import tabulate

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("DELETE FROM Student WHERE Department_ID = 4")

cursor.execute("DELETE FROM Department WHERE Department_ID = 4")

conn.commit()

df = pd.read_sql_query("SELECT * FROM Department", conn)

print("Department Table After Deletion")
print(tabulate(df, headers="keys", tablefmt="grid", showindex=False))

conn.close()
```

Output:

```
Department Table After Deletion
+-----------------+-------------------+
|   Department_ID | Department_Name   |
+=================+===================+
|               1 | CSE               |
+-----------------+-------------------+
|               2 | ECE               |
+-----------------+-------------------+
|               3 | EEE               |
+-----------------+-------------------+
```

Question 7: Display the Student's Name, Student's Department and CGPA if the Student's CGPA is Greater than 8:

Algorithm:

1. Import the required libraries (sqlite3, pandas, and tabulate).
2. Connect to the SQLite database (college.db).
3. Write an SQL query to join the Student and Department tables.
4. Filter the records where the student's **CGPA is greater than 8** using the WHERE clause.
5. Execute the query.
6. Store the result in a Pandas DataFrame.
7. Display the records in a table with rows and columns.
8. Close the database connection.

Code:

```
!pip install tabulate

import sqlite3
import pandas as pd
from tabulate import tabulate

conn = sqlite3.connect("college.db")

query = """
SELECT Student.Name,
Department.Department_Name,
Student.CGPA
FROM Student
JOIN Department
ON Student.Department_ID = Department.Department_ID
WHERE Student.CGPA > 8
"""

df = pd.read_sql_query(query, conn)

print("Students with CGPA Greater Than 8")
print(tabulate(df, headers="keys", tablefmt="grid", showindex=False))

conn.close()
```

Output:

```
Students with CGPA Greater Than 8
+--------+-------------------+--------+
| Name   | Department_Name   |   CGPA |
+========+===================+========+
| Arun   | CSE               |    8.9 |
+--------+-------------------+--------+
| Rahul  | CSE               |    9.2 |
+--------+-------------------+--------+
| Sneha  | EEE               |    8.4 |
+--------+-------------------+--------+
```